Red

Example: Solve $3x + 2 = 14$

The inverse of $+2$ is -2

$x + 2 = 14$

$\boxed{-2}$ $\boxed{-2}$

$3x = 12$

$\boxed{\div 3}$ $\boxed{\div 3}$

The inverse of $x3$ is $\div 3$

$x = 4$

1) $2x + 5 = 25$

$\boxed{-5}$ $\boxed{-5}$

$2x = 20$

$\boxed{}$ $\boxed{}$

$x = 10$

2) $5x + 1 = 11$

$\boxed{-1}$ $\boxed{-1}$

$5x = 10$

$\boxed{}$ $\boxed{}$

$x =$

3) $3x + 10 = 19$

$\boxed{-10}$ $\boxed{-10}$

$3x =$

$\boxed{}$ $\boxed{}$

$x =$

4) $4x + 7 = 23$

$\boxed{-7}$ $\boxed{-7}$

$=$

$\boxed{}$ $\boxed{}$

$=$

mrtonguemaths

Amber

1) $6x + 3 = 15$

$\boxed{-3}$ $\boxed{-3}$

$=$

$\boxed{}$ $\boxed{}$

$=$

2) $7x - 4 = 17$

$\boxed{+4}$ $\boxed{+4}$

$=$

$\boxed{}$ $\boxed{}$

$=$

3) $11x - 5 = 72$

$\boxed{}$ $\boxed{}$

$=$

$\boxed{}$ $\boxed{}$

$=$

4) $8x + 10 = 74$

$\boxed{}$ $\boxed{}$

$=$

$\boxed{}$ $\boxed{}$

$=$

5) $4x - 9 = 27$

$=$

$=$

6) $12x + 4 = 40$

mrtonguemaths

Green

Solve:

1) $5x + 7 = 57$

2) $3x + 2 = 11$

3) $10x - 7 = 63$

4) $9x + 4 = 40$

5) $7x - 12 = 9$

6) $6x + 8 = 80$

7) $9x - 11 = 70$

8) $12x - 20 = 40$

9) $6x + 6 = 66$

10) $13x - 7 = 45$

11) $11x + 32 = 153$

12) $16x - 17 = 79$

Extension

1) A rectangle has width 52 units and **perimeter** 200 units. Find the lengths of the rectangle.

Draw a diagram first!

---

2) Machine A, Machine B and Machine C are three different models of balloon stamping machines.

In one week, Machine B stamps four times as many balloons as Machine A.

Machine C stamps 400 more balloons than Machine A.

All three machines working together stamp 20,200 balloons in one week.

Find the number of balloons stamped by each machine in one week.

If Machine A stamps x number of balloons, how many does Machine B and C stamp in algebraic terms?

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$x =$

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$\boxed{-10}$ $\boxed{-10}$

$3x =$

$\boxed{}$ $\boxed{}$

$x =$

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$\boxed{}$ $\boxed{}$

$x =$

3) $3x + 10 = 19$

$\boxed{-10}$ $\boxed{-10}$

$3x =$

$\boxed{}$ $\boxed{}$

$x =$

4) $4x + 7 = 23$

$\boxed{-7}$ $\boxed{-7}$

$=$

$\boxed{}$ $\boxed{}$

$=$

mrtonguemaths

Amber

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$\boxed{-3}$ $\boxed{-3}$

=

=

2) $7x - 4 = 17$

$\boxed{+4}$ $\boxed{+4}$

=

=

3) $11x - 5 = 72$

=

=

4) $8x + 10 = 74$

=

=

5) $4x - 9 = 27$

=

=

6) $12x + 4 = 40$

Amber

1) $6x + 3 = 15$

$\boxed{-3}$ $\boxed{-3}$

=

=

2) $7x - 4 = 17$

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3) $11x - 5 = 72$

=

=

4) $8x + 10 = 74$

=

=

5) $4x - 9 = 27$

=

=

6) $12x + 4 = 40$

mrtonguemaths

Answers Red:

1) ÷2, ÷2   2) x = 2   3) x = 3   4) x = 4

Amber:

1) x = 2   2) x = 3   3) x = 7   4) x = 8   5) x = 9   6) x = 3

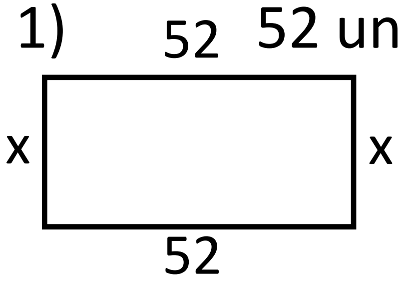
Green:

1) x = 10   2) x = 3   3) x = 7   4) x = 4   5) x = 3   6) x = 12   7) x = 9

8) x = 5   9) x = 10   10) x = 4   11) x = 11   12) x = 6

Extension:

1) 52 units x 48 units



2) A) x balloons   B) 4x balloons   C) x + 400 balloons

x + 4x + x + 400 = 20200   => x = 3300

A = 3300, B = 13200, C = 3700